MI&CT ESERVICE NOTE
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This short-term informational note is invalid 6 months after the issue date.

Date : Oct, 15 2007

Subject : **Jedi No-Load KV Test Update & Spit Error Decoding**

Effectivity : Jedi HP-100/80/RT/VCT and Jedi HP-60/BrightSpeed

Details :

SECTION 1: The purpose of this service note is to communicate recent findings about the Jedi No Load Diagnostic. The KV No-Load Diagnostic can be very useful at isolating a problem between the tube and tank. The Service Engineer should be aware of differences in error reporting thresholds and the need to decode spit types before making any conclusions.

Note 1: When using Jedi Diagnostic mode, each spit is reported as a 60-0316 Max Spit error in the results screen and in the gesyslog, so all spits will be immediately apparent for test purposes. Normally in applications mode the 60-0316 Max Spit error is reported only when greater than 20 spits in 100 ms are detected. Review Section 3 Jedi No-Load Theory for details.

Note 2 Always determine the spit type before making conclusions by decoding the error message Z6 data value using the "Spit Type Decode Table (below)".

> i.e. (60-0316)Spit Max error. PPC board has detected the number of tube spits during exposure has reached the limit **Z6,Z7= 4h,0h**

Note 3: **kV No Load Diag 06-0316 Max Spit errors with Z6=08h (08h= KV Regulation spit type)**
These KV Regulation spit types are typically seen in No Load diagnostic mode more frequently than in applications mode due to test parameter changes and unless they are very frequent there is probably no problem.

Note 4: When using Jedi Diagnostic mode, spits are not included in RunTimeStats.

Note 5: Jedi Diagnostic mode allows use of **150 kV** while normal applications mode is limited to 140 kV. When trying to investigate very intermittent spit issues, then 150 kV can be selected to evaluate the system under higher than normal kV stress. **Please keep in mind that some spits may be expected at 150 kV that will not impact normal application use. The longer the test is run at 150 kV the more likely a spit will occur.** HP60/BrightSpeed Jedi Diagnostic mode does not have the 150 kV capability, 140 kV is maximum on HP60/BrightSpeed design.

When comparing test results between kV No Load with and without the tube in circuit and kV/Ma Diagnostic always remember to compare results at the same kV selections and consider the amount of time each test was run.

Note 6: Perform a Scan hardware reset when switching between Diagnostics kV No Load and kV/mA Diagnostic and when exiting Diagnostics into Applications mode. This will avoid errors such as (60-0329) Wrong HV capacitance measured; or (60-1409) mAs meter saturated.

SECTION 2: DECODING SPIT ERROR MESSAGE

Jedi HP-100/80/RT/VCT Spit Type Decode Table 1

HP100 Spit Type Decode table

Spit Type HEX (Z6)	NA	NA	NA	kV Regulation (Rise Time)	kV Regulation (Exposure)	Anode to Cathode	Cathode	Anode	
	8	4	2	1	8	4	2	1	Decoded spit description
0 4	0	0	0	0	0	**1**	0	0	Anode to Cathode spit
0 8	0	0	0	0	**1**	0	0	0	kV regulation spit during exposure
1 0	0	0	0	**1**	0	0	0	0	kV regulation spit during kV rise
Examples									
0 C	0	0	0	0	**1**	**1**	0	0	kV Reg (exposure) + Anode to Cathode
1 4	0	0	0	**1**	0	**1**	0	0	kV Reg (rise time) + Anode to Cathode

HP100 systems with Monopolar tubes (i.e. Hercules) have a grounded Anode and will report only these spit types

Jedi HP-60/BrightSpeed Spit Type Decode Table 2

HP60 Spit Type Decode table

Spit Type HEX (Z6)	NA	NA	NA	kV Regulation (Rise Time)	kV Regulation (Exposure)	Anode to Cathode	Cathode	Anode	
	8	4	2	1	8	4	2	1	Decoded spit description
0 1	0	0	0	0	0	0	0	**1**	Anode side spit
0 2	0	0	0	0	0	0	**1**	0	Cathode side spit
0 4	0	0	0	0	0	**1**	0	0	Anode to Cathode spit
0 8	0	0	0	0	**1**	0	0	0	kV regulation spit during exposure
1 0	0	0	0	**1**	0	0	0	0	kV regulation spit during kV rise
Examples									
0 C	0	0	0	0	**1**	**1**	0	0	kV Reg (exposure) + Anode to Cathode
1 4	0	0	0	**1**	0	**1**	0	0	kV Reg (rise time) + Anode to Cathode

Error Examples:

60-0316 Max Spit error with Spit Type "kV Regulation" >> Typically ignore these
Wed Sep 19 13:12:24 2007 Host : Jedi Ermes # : 260118213
Exception Class : Abort Severity : Primary File : gen_exec.c Line# : 2376
Function : No System Function Reported Scan Type : Static Scan: 50327/1/1
(60-0316)Spit Max error. PPC board has detected the number of tube spits during exposure has reached the limit (see theory of operation, software section). **Z6,Z7= 8h,0h**

60-0316 Max Spit error with Spit Type "Anode to Cathode" >> Single Anode to Cathode Spit
1190363894 0 1 Fri Sep 21 08:38:14 2007 260118213 4
orp orp_subsys gen_exec.c 2376
Fri Sep 21 08:38:12 2007 Host : Jedi Ermes # : 260118213
Exception Class : Abort Severity : Primary File : gen_exec.c Line# : 2376
Function : No System Function Reported Scan Type : Static **Scan: 50479/1/1**
(60-0316)Spit Max error. PPC board has detected the number of tube spits during exposure has reached the limit (see theory of operation, software section). **Z6,Z7= 4h,0h**

Error Examples:

Switching from Jedi No Load Diagnostic to kV/mA Diagnostic or Applications without a Scan HW reset
1190207867 0 1 Wed Sep 19 13:17:47 2007 260118289 4
orp orp_subsys gen_exec.c 2376
Running diagnostic: kV and mA Test: Xray Functional Test
Wed Sep 19 13:17:46 2007 Host : Jedi Ermes # : 260118289
Exception Class : Abort Severity : Primary File : gen_exec.c Line# : 2376
Function : No System Function Reported Scan Type : Static Scan: 50327/1/1
(60-0329) Wrong HV capacitance measured. JEDI must be reset after KV Diags are performed.

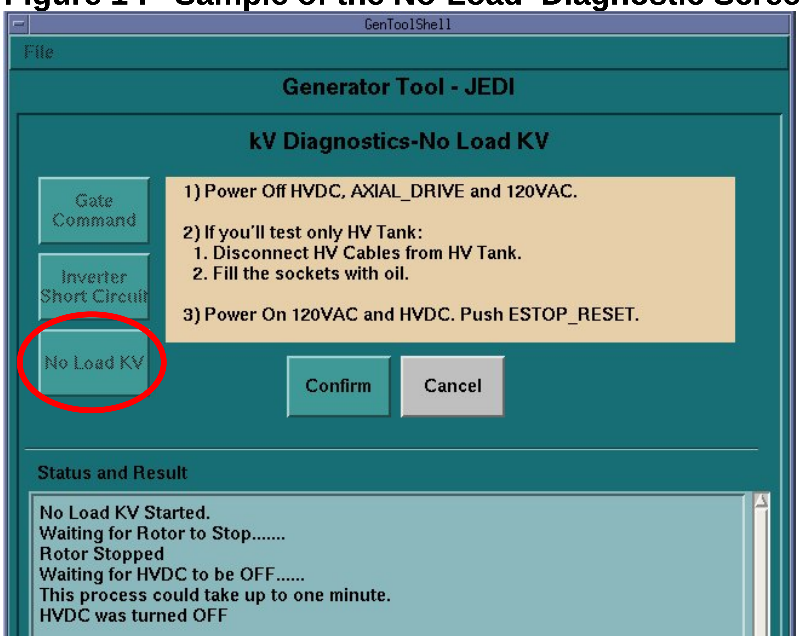
Switching from Jedi kA/mA Diagnostic to Jedi No Load kV Diagnostic without a Scan HW Reset
09/19/2007 16:48:50 260118232 4 orp jedi_subsys gen_exec.c 2376 No System Function Reported Abort Severity : Primary Static Scan: 50329/1/1
(60-1409)mAs meter saturated. A check is done on mAs counter operation at the beginning of exposure and found the mAs meter with unrealistic value. └ Z6,Z7= 0h,0h

SECTION 3: Jedi No-Load Theory: When running the KV No-Load Diag the spit detection sensitivity is turned up to abort on every single Spit. No conclusion should be made solely based on this KV No-Load test errors.

The purpose of the No Load kV test is to verify that the HV power inverter and HV tank are working properly. This test is used when the X-Ray Generation subsystem becomes difficult to diagnose. This test allows for the separation of the HV tank and X-Ray tube by operating with or without the HV candlestick plugged into the HV tank. Thus, this test requires manual operation. A summary of the procedure is provided below: see Service manual for details.

- Run the test via software with HV Candlestick connected. If test passes, the test will not fail when HV cable is disconnected, thus the test is finished.
- Remove HV Candlestick from tank receptacles
- Fill tank receptacle with oil
- Run KV No Load diagnostic

Figure 1 : Sample of the No-Load Diagnostic Screen:



PQR #: 13104026 No Load kV test

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